BOON SOP: THE UNBROKEN DEBT · បុណ្យសុខ៖ បំណុលមិនដាច់

Production document

Development Plan

Scope, roadmap, technical architecture, game systems, assets, resources and open questions for a solo developer.

Engine	Babylon.js 9 (TypeScript) · WebGPU only
Team	Solo developer
First milestone	Episode 1 release (about 35 weeks)
Version	2.1 · October 2026
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Contents

01 **Key Decisions**

02 **Scope and Roadmap**

- The honest size of the project
- Phases and exit criteria
- Episodes 2–5
- Solo cut list

03 **Technical Architecture**

- Stack
- Repository layout
- Core architecture
- Rendering
- Performance budgets
- Save system
- Browser support
- Input and accessibility

04 **Game Systems**

- System matrix
- Fail state
- Cross-episode tracking
- Entity catalogue

05 **Assets and Resources**

- Tools
- Episode 1 asset list
- Asset sources
- Pipeline
- Learning
- Cultural research
- Budget

06 **Open Questions**

01 Key Decisions

This plan turns the proposal and the five episode specifications into a plan one person can carry out. These decisions are fixed unless deliberately revisited.

Decision	Choice	Reason
Team	Solo developer	Sized for one person; art and audio come from libraries plus targeted custom work.
Engine	Babylon.js (TypeScript)	Mature WebGPU renderer, built-in Havok physics, spatial audio, inspector, no editor lock-in.
Target hardware	Any laptop or desktop with WebGPU; phones and tablets not supported	Minimum: integrated graphics (Iris Xe, Radeon 680M, Apple M1, Snapdragon X) on Low. Recommended: RTX 2060 / RX 5700 / M1 Pro on Medium at 1080p. High-end: RTX 3070 / RX 6800 / M2 Pro on High or Ultra. A Low preset and dynamic resolution roughly double the audience for 2–3 weeks of work; phones would cost 2–3 months (touch controls, memory, heat), so a mobile version waits until after release. No WebGL2 fallback, so effects can use WebGPU fully and testing is simpler.
Assets	Libraries + custom Blender work	Generic props from CC0 or paid libraries; Khmer-specific hero assets made custom.
First milestone	Episode 1 released on its own	Proves the technology, tone and pipeline before committing to four more episodes.
Episode length	1.5–2.5 hours each, 8–10 hours total	Short, dense episodes (5–6 puzzles, 3–4 escalating encounters, one optional area, one key choice) are better for players and achievable alone.

Versions checked (October 2026)

Package	Version
`@babylonjs/core`, `@babylonjs/loaders`	9.31.0
`@babylonjs/havok`	1.3.14
`vite`	8.3.4
`typescript`	6.0.3 (TypeScript 7 is not yet supported by ESLint tooling)
Node.js (installed)	26.8.2

Pin exact versions in `package.json` when the project is created, and upgrade deliberately.

02 Scope and Roadmap

2.1 The honest size of the project

The original proposal budgeted 11 months for a five-person team, roughly 55 person-months. With the shorter, denser episodes chosen in v3.0 (8–10 hours in total instead of 15–20), a solo developer should expect about **2–2.5 years** at 25 hours per week. To keep it achievable:

1. **Ship Episode 1 alone first**, as a free or pay-what-you-want release, to test the idea with real players.
2. **Build every system once** in Episode 1 and reuse it in Episodes 2–5.
3. **Cut cost, not atmosphere.** Darkness, fog, sound and first-person framing hide a lot.

Estimates assume about 25 focused hours per week. Scale them to your real availability.

PHASE 0	PHASE 1	PHASE 2
Pre-production	**Technical spikes**	**Greybox**
3 weeks	4 weeks	4 weeks

PHASE 3	PHASE 4	PHASE 5
Core systems	**Art and audio**	**Polish and release**
7 weeks	12 weeks	5 weeks

Total: about 35 weeks (8 months) to the Episode 1 release. Episodes 2–5 then take about 4 months each. Art and audio take longer because of the high-end quality target.

2.2 Phases and exit criteria

Each phase ends with a checklist. Don't start the next phase until it passes, unless you consciously accept the gap.

Phase 0 · Pre-production — 3 weeks

Task	Output
Review the Story Bible with a cultural advisor; resolve remaining open questions (section 6)	Approved story
Write the Episode 1 beat sheet: every scene, trigger, item and line in order	Done: document 08 (beat sheet and level layout) and 10 (script)
Define the fail state (what happens when Sora is caught)	Done: section 4.2 and document 08
Build a visual reference board: 1930s Battambang houses, funeral tents, lamps, clothing	Reference folder with sources
Write the art bible: palette, lighting rules, post-processing, texture density	Done: document 09

Task	Output
Write the technical design and controls specifications	Done: documents 11 and 12
Install the toolchain (section 5.1)	Working Blender → glb → browser pipeline
Work through the Babylon.js getting-started docs and playground	Comfort with scenes, cameras, lights, materials, loaders

Exit: you can describe all of Episode 1 from start to end without opening the specification.

Phase 1 · Technical spikes

4 weeks

Small throwaway prototypes that answer risky technical questions before they become expensive.

Spike	Question	Pass condition	Status
S1 First-person controller	Does movement feel right on raised floors, stairs and ladders?	Walk, crouch, climb stairs, head-bob; no snagging on 1 cm floor gaps	Passed
S2 Lamp lighting	Can the lighting target run on recommended spec?	60 fps at 1080p on Medium (RTX 2060 / M1 Pro class) with the lamp, 2 live-shadowed flames, 32+ unshadowed lights, SSAO and volumetric fog	Passed (Chrome)
S3 Asset pipeline	Can a Blender model reach the browser compressed?	.blend → .glb → Draco/meshopt + KTX2 → loads with correct materials	Passed
S4 Spatial audio	Can players locate an unseen enemy by sound?	Eyes closed, you can point to a footstep source	Not started
S5 Diegetic sketchbook	Can a 3D book show readable, updatable pages?	Open, flip, new note appears; readable at 1080p	Not started

Spike	Question	Pass condition	Status
S6 Save and load	Does state survive a refresh and export as a code?	Save, reload, restore; export and import a save code	Not started
S7 Atmosphere	Can the 1930s look be done cheaply?	Grain, vignette, aberration, sepia grade and fog within budget	Not started
S8 WebGPU browsers	Does the build run in every WebGPU browser?	Same scene runs in Chrome, Edge, Safari (macOS 26) and Firefox (Windows); unsupported browsers and devices show the requirements page	Chrome passed; Safari, Firefox and Edge not tested

Exit: all spikes pass, or the design is adjusted (for example, fewer shadowed lights).

FINDINGS SO FAR (OCTOBER 2026)

S1 controller	A bot walks the whole test course (stilt-house floor with 1 cm plank gaps, 20 cm and 25 cm stairs, a 6 cm threshold, a low passage and a granary ladder) with no snags. Havok's character controller can't step up stairs at walking pace, so stairs get hidden ramp colliders (`col_` meshes in Blender) and step-up is off; the rounded capsule rides over edges up to about 10 cm. Ground snapping runs on flat floors only, so it never pushes the capsule into a ramp. Ladders are trigger volumes (`ladder_`); crouching checks headroom before standing up.
S2 lighting	On the reference MacBook Pro M1 Pro, Medium at 1080p averages 62.5 fps (from 15 fps in the first build); Low runs at about 100 fps at 900p. The design was adjusted to fit: 2 flames cast live shadows on Medium, and their shadow maps are cached and redrawn only when something moves nearby; small props don't cast shadows; flame lights only affect meshes in range; static meshes are merged by material in 3 m cells; fog and SSAO blur run at half resolution. Two engine pitfalls cost the most: a pre-pass setting that forced the fog to full resolution, and shadow caster lists rebuilt each frame, which recompiled every material every frame.

S3 assets	A test corner of a stilt house, built by a Blender script, goes .blend → .glb (`npm run export`) → meshopt + KTX2 (`npm run assets`) and loads in 0.25–0.35 s. The file shrinks from 2.8 MB to 1.6 MB, and the textures from about 50 MB of GPU memory to about 10 MB: colour maps (ETC1S) arrive as ETC2 and normal and ORM maps (UASTC) as ASTC on the M1 Pro. Materials keep colour, normal and packed ORM maps, with occlusion read from the ORM red channel. A first `LevelBuilder` turns `col_` , `ladder_` , `spawn_` , `patrol_` and `interact_` nodes into colliders, a ladder, a spawn point and a route, and the S1 bot walks the route (stairs, deck, an earth mound, ladder) with no snags. Problems fixed on the way: `prune` deleted every empty node (spawns and patrols) unless told to keep leaves; the CLI splits `{a,b}` slot globs, so the documented commands failed; gltf-transform 4 needs KTX-Software's `ktx` , not `toktx` ; Blender exported every material double-sided; and Babylon fetched its meshopt and KTX2 decoders from its CDN, which now ship with the game instead.
S8 Chrome and the S2 1% low	In standalone Chrome 155 on the M1 Pro, S1 (bot: 81 s, no snags) and S3 (all checks; loads in 0.7 s warm from the dev server) pass, and the S2 benchmark on Medium at 1080p averages 60 fps with a 1% low of 48–50 fps: in 1,200 frames, 3 take over 20 ms and none over 25 ms. The 1% low of about 30 fps seen earlier comes from the in-app development browser, not the game, so S2 passes. Benchmark in a standalone browser from now on.
Still open	S8: Safari (macOS 26) and Firefox on this Mac, and Edge on Windows, before Phase 2 starts. S2 is not yet measured on an RTX 2060 or on integrated graphics, and GPU timing isn't available on this Mac, so dynamic resolution works from frame times.

Spikes run in the dev build: `?spike=s1` (add `&auto=1` for the bot), `?spike=s2` (add `&bench=1` for a timed benchmark, `&quality=low|medium|high|ultra` for a preset) and `?spike=s3` (add `&auto=1` for the bot, `&file=assets/…glb` to inspect another asset). F3 shows the performance overlay.

Phase 2 · Episode 1 greybox

4 weeks

Build the whole episode from grey boxes: no final art, but every room at the right scale and every trigger, puzzle, enemy path, line of text and the dawn key choice in place. Target: 1.5–2 hours for a first-time player.

Exit: a friend can play Episode 1 start to finish, and it is scary even with boxes. If it isn't, fix the pacing now; art will not save it.

Phase 3 · Core systems

7 weeks

Turn the spike code into reusable systems (section 4): interaction, inventory, sketchbook, lamp and resources, stealth perception, entity AI, scripted events, saves, settings and accessibility. Add

automated tests for game logic.

Exit: adding a new item, note or trigger means editing data, not engine code.

Phase 4 · Art and audio

12 weeks

Weeks	Focus
1–3	Hero architecture: stilt house, funeral tent, under-house area, shed
4–7	Props (library assets retextured, custom Khmer props, betel tray set) and characters (Yay Mey, Vuthy, the corpse, the entity)
8–9	Lighting bake, post-processing, rain and wind effects
10–11	Sound: ambience, foley, horror cues, entity sounds, music
12	Performance pass: draw calls, polygons, texture memory, load size

Exit: meets the performance budgets (section 3.5) on both reference machines.

Phase 5 · Polish, test, release

5 weeks

- Five or more outside playtesters; watch without helping and note where they get lost or bored.
- Cultural review by at least one Cambodian advisor.
- Browser testing: Chrome, Edge, Firefox and Safari on Windows and macOS.
- Store or landing page, trailer capture, release.

Exit: Episode 1 is public. Collect feedback before starting Episode 2.

2.3 Episodes 2–5

Episode	New systems	Reused	Estimate
2 · Whispering Pagoda	Lamp shutter, surface noise, breath, distractions, mirror beams, playable flashback (alternate hands and pace), timed escape	Everything from Episode 1	3–4 months
3 · Drowned Village	Daylight lighting and simple NPCs, rowboat, wetness, safe-zone lights, water-only enemy	Fuses reuse the key-fragment pattern	3.5–4 months
4 · Rubber Plantation	Compass, fog loop, voice-mimic taboo, companion AI (follow, wait, carry light), furnace sequence	Gears reuse inventory slots	3–4 months

Episode	New systems	Reused	Estimate
5 · Sunken Sanctuary	Hub with escalation, boss ritual, chant choices, ending resolver	Each trial reuses an earlier system	3.5–4 months

2.4 Solo cut list

If time runs short, cut from the top. None of these damages the core experience.

1. Spoken dialogue → written text, with Khmer voice for key lines only.
2. Full-body characters → Sora is hands-only; NPCs mostly seated, still or in silhouette.
3. Many unique props → fewer props, varied with material tints and decals.
4. Physics-driven debris in escapes → scripted animation.
5. Rowboat physics (Episode 3) → boat on a path with limited steering.
6. Daylight market crowd (Episode 3) → a few static vendors and ambient sound.

> **AVOID AS A SOLO DEVELOPER**
>
> - Building a level editor: use Blender as the level editor and export metadata with custom properties.
> - Writing engine features Babylon.js already has (physics, audio, picking, GUI, post-processing).
> - Starting Episode 2 content before Episode 1 is released.
> - Polishing art in a room whose gameplay is not final.

03 Technical Architecture

3.1 Stack

Layer	Choice	Notes
Language	TypeScript (strict)	Babylon.js itself is written in TypeScript
Build	Vite	Fast dev server; static build output
Engine	`@babylonjs/core` (ES modules)	Import only what is used to keep the bundle small
Loaders	`@babylonjs/loaders` (glTF 2.0)	Draco/meshopt and KTX2 decoders loaded on demand
Rendering	WebGPU only	Check support at startup; show the requirements page if WebGPU is missing
Physics	Havok	Character controller, collisions, triggers
Audio	Babylon.js Audio Engine v2	Spatial sound, buses, reverb; unlocked on first click
Interface	Babylon GUI on meshes + dynamic textures	Diegetic sketchbook and gauges; HTML only for menus and loading
Navigation	Recast navmesh plugin	Enemy pathing (check current docs for the package)
Storage	IndexedDB	Save slots and settings
Tests	Vitest; Playwright smoke test	Test game logic, not rendering
Assets	Blender → glTF → glTF-Transform	One command rebuilds all assets
Hosting	Static host (Cloudflare Pages, Netlify, GitHub Pages, itch.io)	Check per-file size limits
Source control	Git + Git LFS	LFS for .blend, .wav and other binary sources

3.2 Repository layout

```
/
├── docs/                Design documents, PDFs, sources
├── art-source/          .blend, textures, raw audio (Git LFS)
├── tools/               Asset and document build scripts
├── public/assets/
│   ├── shared/          Lamp, hands, satchel, sketchbook, common sounds
│   └── ep1/             Episode 1 compressed models, textures, audio
├── src/
│   ├── main.ts          Boot: device check, create engine, load menu
│   ├── core/            Engine wrapper, scene manager, loader, event bus
│   ├── player/          Controller, camera, interaction, hands
│   ├── systems/         Inventory, sketchbook, lamp, stealth, save, settings
│   ├── ai/              Perception, state machines, entity types
│   ├── audio/           Audio setup, buses, ambience zones, cues
│   ├── render/          Device check, quality presets, dynamic resolution, post-processing
│   ├── ui/              Diegetic widgets, HTML menus
│   ├── episodes/ep1/    Scene setup, scripted events, puzzles, data
│   ├── data/            Shared item, note and dialogue definitions
│   ├── debug/           Performance overlay and benchmark
│   └── spikes/          Phase 1 technical spikes
└── tests/
```

3.3 Core architecture

Principle: systems are generic; episodes are data plus a few scripts.

Part	How it works
Event bus	Systems never call each other directly. The lamp announces `lamp:extinguished`; the AI listens and lowers visibility. This keeps systems reusable.
Game flags	One key-value store holds all story progress (for example `ep1.gramophone.solved`). Saves serialise it; Episode 5 reads flags from Episodes 1–4.
Scripted sequences	Written as readable async functions (walk to, play sound, wait for look), so no visual scripting tool is needed.
State machines	Entities and puzzles use small explicit state machines (idle → patrol → investigate → chase → lose track) in plain TypeScript.
Blender as level editor	Triggers, spawn points, patrol nodes, hiding spots and audio zones are empties with custom properties, exported to glTF and turned into game objects by the loader.
Area streaming	Each episode is split into areas loaded and unloaded as asset containers. Shared assets stay loaded; unloaded areas dispose their meshes, materials and textures.

3.4 Rendering

Feature	Approach
Lamp light	Point light with high-resolution soft shadows (the hero light).
Shadowed flames	Up to 4 additional shadow-casting flame lights in key moments (the tent, the altar, the ritual).
Candles and small flames	Dozens of unshadowed point lights with flicker, using Babylon 9's clustered lighting; emissive flames and bloom.
Static lighting	Baked lightmaps from Blender for architecture, combined with real-time SSAO and contact shadows.
Reflections	Screen-space reflections on wet mud, puddles and floodwater (Episodes 1 and 3).
Fog	Volumetric fog and light shafts (WebGPU compute), essential for Episodes 2 and 4.
Post-processing	Temporal anti-aliasing, grain, chromatic aberration, vignette, bloom, depth of field in memories, and colour grading (a sepia look for Episode 1).
Special materials	Node Material Editor for the entity shadow, wet surfaces, blood on the map and glowing runes.
Quality presets	Low (integrated graphics) / Medium (recommended spec) / High / Ultra. The device check picks a starting preset from the graphics adapter; dynamic resolution lowers the render scale when frames run over budget. Gameplay is identical on every preset.

3.5 Performance budgets

Budget	Target	How to check
Frame rate	60 fps at 1080p on recommended spec (Medium) · 60 fps at 1440p on high-end (High/Ultra) · Low on integrated graphics aims for 60 fps, with dynamic resolution down to 60% of 900p	The S2 benchmark (`&bench=1`), the F3 overlay and browser GPU profiling
Draw calls	About 1,000 or fewer	Inspector statistics; instancing and WebGPU snapshot rendering
Visible triangles	Up to about 3 million	Inspector; LODs on everything beyond hero range
Download	First playable area under 150 MB; full episode up to about 600 MB, streamed and cached	Browser network tab with cache disabled
GPU memory	About 3 GB per area at Ultra, about 2 GB at Medium, about 1 GB at Low	KTX2 everywhere; 1K, 2K and 4K texture tiers

Budget	Target	How to check
JavaScript bundle	3 MB or less (gzipped)	Vite build report

The reference machine is a MacBook Pro M1 Pro (recommended spec, Medium), benchmarked every week with other apps closed. Before release, check Low on an integrated-graphics laptop and Medium on an RTX 2060-class PC, borrowed, bought used or through playtesters.

3.6 Save system

What is saved	Episode, area, checkpoint, flags, inventory, sketchbook entries, resource levels, settings and ending counters (taboos broken, lore notes found).
When	Fixed checkpoints and episode completion; no free saving, which keeps the tension.
Where	IndexedDB, three slots, versioned, with migrations for future changes.
Save codes	JSON → compressed → URL-safe base64 with a checksum, for moving progress between devices and into the next episode.
Security note	Browser-side encryption cannot stop a determined player because the key ships with the game. Save codes are tamper-evident, not secret, which is fine for a single-player game.

3.7 Browser support

Browser	Expected path
Chrome, Edge	WebGPU
Safari	WebGPU on macOS 26+; older macOS not supported
Firefox	WebGPU on Windows and Apple Silicon Macs; other platforms not supported
No WebGPU, or a software renderer	Requirements page with the system requirements and supported browsers
Integrated graphics	Supported on Low
Phones and tablets (including iPad)	Requirements page; not supported in Episode 1

3.8 Input and accessibility

- Keyboard and mouse first; gamepad support through the Gamepad API.
- Rebindable keys, mouse sensitivity, invert Y, field-of-view slider.
- Subtitles and captions for important sounds, with a visual-cue option for deaf and hard-of-hearing players.
- Toggles for head-bob, film grain and chromatic aberration (motion sickness and comfort).
- Brightness calibration screen.

- English and Khmer text from the start, with all strings stored in data files.

04 Game Systems

4.1 System matrix

Every mechanic from the five episodes, with the episode that first needs it. Build each system once and extend it.

System	Ep	Reused in	Build notes
First-person controller	1	All	Havok character controller; crouch lowers camera and noise
Interaction raycast	1	All	Ray from screen centre; feedback on the hands, never a HUD crosshair
Inventory (satchel)	1	All	Data-driven items; satchel opens in front of the camera
Sketchbook	1	All	3D book; pages rendered to texture; entries unlocked by flags
Kerosene lamp	1	All	Oil gauge is a mesh on the lamp; refills are pickups
Matches	1	5	Count shown on the matchbox
Timed upkeep (incense)	1	5	Generic decaying resource: timer, warnings, consequence
Hiding spots	1	2, 4	Trigger volume + lamp off = hidden
Perception: sight	1	All	View cone + line-of-sight ray; lamp light raises visibility
Perception: hearing	1	All	Noise events with loudness and radius
Moves when unwatched	1	5	Checks camera view and occlusion
"Do not turn around"	1	4, 5	Tracks camera rotation against a reference direction
Scripted sequences	1	All	Async scripts reading and writing flags
Puzzle framework	1	All	Slots, dials, combinations
Ritual sequence	1	3, 4, 5	Ordered steps with a timer and failure handling
Checkpoints and saves	1	All	See section 3.6
Night count	1	All	Seven circles in the sketchbook; one filled per episode
Choices and counters	1	All	Respect, knowledge, Vuthy; stored in flags and save codes
Lamp shutter	2	3, 4, 5	Lamp states: open / shuttered / out

System	Ep	Reused in	Build notes
Surface noise	2	All	Material tags on ground meshes
Hold breath	2	5	Short timed action with recovery
Throwable distractions	2	3, 4	Physics throw + noise event on impact
Mirror light beams	2	5	Reflected raycasts between rotatable mirrors
Timed escapes	2	3, 4	Scripted hazards with a checkpoint at the start
Playable flashback / memory	2	3 (Sok), 5 (memories)	Swap hands model, speed and post-processing; no fail state
Daylight scene and NPCs	3	—	Second lighting preset; simple talk-to NPCs
Rowboat	3	—	Simplest viable version first
Wetness and temperature	3	—	Drains in water, recovers near heat
Safe-zone lights	3	4	Area that repels an entity type
Sealing windows	3	—	Puzzle slots that change a room's flooding
Water-only enemy	3	—	Separate navmesh of water surfaces
Compass	4	5	Needle follows world north
Fog loop	4	—	Wrong turns quietly loop the player
Voice-mimic taboo	4	—	Turning or noise counts as answering
Furnace and valves	4	—	Resource-juggling ritual variant
Crawling	4	5	Extends crouch; forced slow speed
Companion AI	4	3 (Pich, simplified)	Follow, wait, carry a light; never speaks from behind the player. Pich reuses a simple "lead to a point" version
Telephone interlude	4	—	Short dialogue scene with one unscored choice
Hub with trials	5	—	Flags per trial; difficulty scales with trials completed
Chant choices	5	—	Choices drawn in the sketchbook
Boss ritual	5	—	Ritual sequence + entity attacks
Ending resolver	5	—	Three endings from chant errors, respect and Vuthy (section 4.3)

4.2 Fail state

The episode specifications never said what happens when Sora fails. Recommended:

- Caught → short scare, fade to black, a page torn from the sketchbook, restart at the last checkpoint.

- No health bar. Episode 5's shadow damage is shown by darkening and sound, with two or three hits before failure.
- Running out of oil doesn't kill; it makes the dark more dangerous.

4.3 Cross-episode tracking

Counter	Example sources
Respect (0–9)	Ep 1: betel tray, incense (+2) · Ep 2: offerings, honest memory, names at dawn (+3) · Ep 3: mother freed (+2) · Ep 4: no taboo broken, boundary stones (+2)
Knowledge	About 5 lore notes per episode; they confirm the true chant lines in Episode 5
Vuthy	Journal shared (Ep 1, dialogue only) · saved (Ep 4) · forgiven (Ep 5)
Chant errors	Wrong verses in Episode 5, Phase 2
Endings	3+ chant errors → Bound to the Soil · otherwise respect ≥ 6 + Vuthy saved + forgiven → The Unbroken Freedom · otherwise → The Last Payment

4.4 Entity catalogue

Entity	Ep	Senses	Behaviour	Counter-play
Neak Ta shadow	1–5	Sight + hearing; stronger each episode	Mostly scripted; roams set areas	Darkness, hiding, rituals
Neak Ta shadow forms	1, 4, 5	Mostly hearing (Ep 1); sight of bright light (Ep 4)	House walker and under-house patrol (Ep 1), manor shadows (Ep 4), moving statues (Ep 5); one entity with different settings	Quiet movement, lamp off, shaded light, watching
Pich (Mrenh Kongveal)	3, 5	—	Leads Sora to safe routes; never hostile	Follow him (optional)
Pret	2, 5	Hearing only	Patrol stupa paths; drawn to offerings	Crouch, soft ground, breath, distractions
Chhlong Tonle	3	Senses the player in water	Moves only through water	Stay dry, safe-zone lights
Khmaoch Prey	4	Looking back or replying	Mimics voices from behind	Walk on, don't turn, stay silent
Neak Ta Guardian	5	Full awareness	Area attacks, tendrils	Complete the ritual while evading

05 Assets and Resources

5.1 Tools

Purpose	Tool	Cost	Notes
Code editor	VS Code	Free	
3D, UVs, baking, level layout	Blender (latest LTS)	Free	Not installed yet
Asset compression	glTF-Transform CLI	Free	Draco/meshopt, KTX2, resizing
KTX2 encoder	KTX-Software (Khronos)	Free	Used by glTF-Transform
Texture painting	Krita · Substance 3D Painter	Free / paid	ArmorPaint, Material Maker as open alternatives
Image editing	GIMP · Photopea	Free	
Audio editing	Audacity · Reaper	Free / ~US$60	
Recording	USB microphone, quiet room	—	Foley is cheap to record yourself
Binary version control	Git LFS	Free	Not installed yet
Babylon tools	Playground, Sandbox, Node Material Editor, Inspector	Free	Browser-based

5.2 Episode 1 asset list

Source: **C** custom (Blender) · **L** library asset, retextured · **K** kitbash · **R** record yourself.

ENVIRONMENTS

Asset	Source	Priority	Notes
Phnom Penh studio interior	K	High	Small room; reuse furniture later
Village path and banana grove	K	Medium	Instanced banana plants; mostly darkness and rain
Courtyard ground, mud, puddles	C	High	Wet mud material is a key visual
Teak stilt house (exterior, hall, Sok's room)	C	Critical	Modular kit: planks with gaps, walls, posts, stairs, roof, shutters
Under-house area and storage shed	C	High	Reuses house pillars
White funeral tent	C	Critical	Bamboo frame; cloth with wind animation

PROPS, CHARACTERS AND AUDIO

Asset	Source	Notes
Teak coffin and stage	C	Hero prop; also used in the bad ending
Altar, portrait frame, brass candle holders	C / L	Portrait texture swaps for the "eyes open" scare
Four brass incense burners and sticks	C	Burn-down shader on sticks
Protective white thread	C	Snaps in two
Oil lamps, candles	C / L	Emissive flame card + flicker light
Clay jars, bamboo benches, mats	C / L	2–3 variants, instanced
Gramophone with speed dial	L + edits	Adjust to the period
Desk, pens, palm-leaf manuscripts	L / C	
Iron-bound chest, loose planks, wardrobes	C / L	Wardrobes are hiding spots
Telephone, wall clock, drafting table	L	Prologue
Player items: lamp, matchbox, satchel, sketchbook, key fragments, Bronze Token	C	Seen up close all game; most detail
First-person hands	C / L	No full body needed
Yay Mey	L + custom	White mourning clothes; few animations
Vuthy	L + custom	1930s town suit; standing and talking only in Episode 1
Betel tray set (leaves, areca, lime pot, tray)	C	Puzzle props; check against reference
Uncle Sok's corpse	C / L	Lying pose only
Neak Ta entity and shadow roamer	C	Mostly shader silhouette plus a detailed brass mask
Ambience beds	L + R	Rain, thunder, insects, frogs, wind, roof tiles
Foley	R	Footsteps on teak, dirt and mud; cloth; pages; matches; creaks
Horror cues	R + design	Backward chant, clock, thread snap, dragging steps, whispers, static
Music	Commission / licence	Sparse Pinpeat-inspired tones and drones
Voice	Khmer voice actors	Phone call and key lines, English subtitles

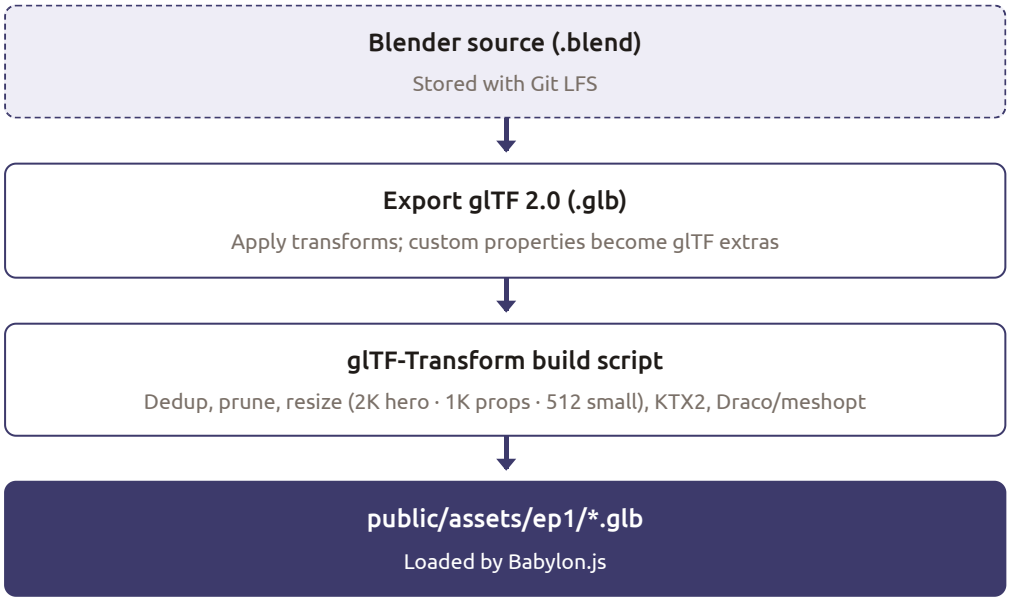
5.3 Asset sources

Check the licence of each individual asset and record source, author, licence and link in the asset tracker (`docs/production/asset-tracker.csv`), which becomes the credits list.

Source	What	Licence notes
Poly Haven	Textures, HDRIs, some models	CC0
ambientCG	PBR textures	CC0
Sketchfab	Models, including some Southeast Asian items	Mixed; filter CC0/CC-BY and credit authors
Fab (Epic)	Megascans and marketplace	Megascans no longer free after 2024; check the current licence
Freesound	Sound effects	Per-sound CC licences
Sonniss GameAudioGDC bundles	Large royalty-free SFX packs	Free; check the included licence
Mixamo	Rigging and animations	Free with an Adobe account; check terms
Google Fonts	Khmer fonts: Noto Serif Khmer, Battambang, Moul, Koulen, Preahvihear	Open Font License

> **RULE OF THUMB**
>
> Use library assets for generic objects (furniture, tools, textures) and custom work for anything that makes the game Cambodian: architecture, ritual objects, clothing and the entity. Players notice the second group.

5.4 Asset pipeline



Naming: `ep1_house_wall_a.glb`, `shared_lamp.glb`; Blender empties named `trigger_*`, `spawn_*`, `patrol_*`, `hide_*`, `audio_zone_*` become game objects.

5.5 Learning resources

Topic	Where
Babylon.js fundamentals	doc.babylonjs.com (Getting Started, then Features); playground.babylonjs.com
Help and community	forum.babylonjs.com
WebGPU, physics, audio	doc.babylonjs.com: WebGPU, Physics V2 (character controller), Audio
Shaders without code	nme.babylonjs.com (Node Material Editor)
glTF optimisation	gltf-transform.dev
Blender for games	Blender manual; modular environment and trim-sheet tutorials
Horror design	Play and study *Amnesia: The Dark Descent*, *Fatal Frame*, *Home Sweet Home*, *Pamali*, *DreadOut*; GDC talks on horror pacing and audio
Web performance	Babylon.js "Optimizing your scene"; browser performance tools

5.6 Cultural and historical research

Topic	Sources to start with
Neak Ta and Khmer spirits	Ang Choulean, *Les êtres surnaturels dans la religion populaire khmère* (Paris, CEDORECK, 1986). Neak Ta are usually protective guardians, often represented by stones.
Colonial economy and plantations	Margaret Slocomb, *An Economic History of Cambodia in the Twentieth Century* (NUS Press, 2010; open access via OAPEN); Michitake Aso, "Rubber and Race in Rural Colonial Cambodia (1920s–1954)".
Historical photographs	Gallica (Bibliothèque nationale de France), Archives nationales d'outre-mer (ANOM), École française d'Extrême-Orient (EFEO), Bophana Audiovisual Resource Center.
Funerals, Pchum Ben, offerings	Ethnographic literature plus interviews with elders and monks (with permission).
Pinpeat music	Cambodian Living Arts and Cambodian musicians; commission or license recordings rather than imitating them.
Language	A native Khmer speaker to check every Khmer term and translate in-game text.

> **ADVISOR PLAN**
>
> Find one or two Cambodian advisors in Phase 0 (a family elder, a university contact or a cultural organisation). Schedule reviews of the story outline, the Episode 1 greybox and the release candidate. Credit them in the game.

5.7 Rough budget (Episode 1)

Item	Low	Comfortable
Software	US$0	~US$300
Paid assets	US$0–100	~US$400
Sound and music	US$0	US$500–1,500
Voice acting	US$0	US$200–600
Cultural advisor honorarium	—	US$200–500
Test hardware (used integrated-graphics laptop; RTX 2060-class GPU)	—	US$250–500
Hosting	US$0	US$0–20 / month

06 Open Questions

Found by reading the proposal and all five episode specifications together. Each episode document also lists its own open issues.

6.1 Story and continuity

#	Issue	When	Suggested resolution
Q1	The Silver Betel Box is never destroyed or cleansed, yet Episode 5 needs its "inlay"	Done	Dawn offering at Chea's stupa added in v3.0
Q2	Two versions of the pact: forest altar (Ep 2) vs estate contract (Ep 4)	Done	1910 forest shrine; the estate was built over it in 1923
Q3	A working rubber estate in 1910 is historically early	Done	Resolved by Q2
Q4	Sora's parents are missing; Yay Mey's place in the bloodline is unclear	Done	Kanha was the third payment (1920); father Rith died 1929; Yay Mey married in
Q5	Clinic timeline (1920 vs "decades ago" vs 1925 photo)	Done	Kanha drowned in 1920; Sok returned in 1931; photo dated 1920
Q6	Time between episodes is never stated	Done	Seven-night countdown: Nights 1, 3, 4, 5, 6, 7
Q7	Time mismatches in v1 headers; sunrise at 4:00 AM	Done	Corrected in the v2.0 episode documents
Q8	The Episode 1 map already shows all four locations	Done	Four faint marks, revealed one at a time
Q9	What happened to Kanha's baby in 1920?	Done	Her son Pich died with her and lingers as a gentle Mrenh Kongveal (v3.2)

6.2 Gameplay

#	Issue	When	Suggested resolution
G1	No fail state defined	Done	Checkpoint restart, no health bar (section 4.2)
G2	Ending conditions aren't tracked in Episodes 1–4	Done	Respect, knowledge and Vuthy counters; one key choice per episode (section 4.3)
G3	Checkpoints only, or manual saves too?	Before Ep 1	Checkpoints only
G4	Single difficulty or a story mode?	Before release	A story mode helps players who come for the culture and narrative

#	Issue	When	Suggested resolution
G5	"Hand-crank lantern" in Episode 4	Done	Changed to an oil lantern in v2.0
G6	"Kerosene spotlights" in Episode 3	Done	Changed to fixed wall lamps in v2.0
G7	Pret appear in the Episode 5 Wind trial	Before Ep 5	Fine as a callback; explain in the sketchbook

6.3 Presentation, culture and business

#	Issue	When	Suggested resolution
C1	Neak Ta as the antagonist; they are normally respected protectors	Before Ep 1	A guardian enforcing a broken contract, not evil
C2	Several Khmer terms in v1 need checking (see each episode's glossary)	Each episode	Native speaker review; keep one glossary
C3	"Graphic designer" is a modern job title	Done	"Illustrator" used in v2.0
C4	Language of voices and text	Before Ep 1	Khmer voices; English and Khmer text
C5	The blood offering in Episode 5	Before Ep 5	Review with the advisor; keep it restrained
B1	Is Episode 1 free, pay-what-you-want or paid?	Before release	
B2	Where to publish (itch.io, own site, both)?	Before release	itch.io is the easiest start for a browser game
B3	Licence: the README says MIT for everything	Done	Code under MIT; story, art, audio and documents all rights reserved (LICENSE, LICENSE-CONTENT.md)